GMMDA: Gaussian Mixture Model-Based Generative Domain Adaptive Semantic Segmentation for Remote Sensing Images

Yongkang Hu, Xiaogang Yu, Yupei Wang, and Liang Chen

***Abstract*—Unsupervised domain adaptation (UDA) for semantic segmentation transfers knowledge from an annotated source domain to an unlabeled target domain, reducing the cost of pixel-level annotation. Most existing methods rely on dense discriminative classifiers that estimate pixel-wise class posterior probabilities using a single weight vector per class. However, in remote sensing imagery, intraclass variability often gives rise to multipeaked feature distributions with multiple subclusters, which may not be adequately captured by a single weight vector, leading to intraclass confusion. Moreover, such discriminative classifiers primarily focus on learning class decision boundaries, while the consistency of same-class subclusters across domains is only weakly constrained. Under domain shift, these subclusters may drift inconsistently, thereby aggravating cross-domain structural mismatch. To this end, we propose GMMDA, a Gaussian mixture model (GMM)-based generative domain adaptation method that predicts pixel labels through Bayesian inference based on class-conditional feature likelihoods. Specifically, we fit a class-wise GMM using an online momentum Sinkhorn EM procedure to model class-conditional feature densities, where each Gaussian component corresponds to an intraclass subcluster, enabling a richer characterization of multipeaked distributions. Furthermore, component-aware contrastive learning (CCL) treats Gaussian components as fine-grained semantic References, pulling target-domain features toward their best-matched components using a variance-aware distance that accounts for component scales. In addition, we introduce componentwise distribution alignment (CDA) to align each source component with the corresponding source-guided target local statistics by jointly constraining their means and principal directions, aiming to reduce cross-domain structural mismatch. Experimental results show that GMMDA achieves competitive mean intersection over union (mIoU) scores on the international society for photogrammetry and remote sensing (ISPRS) and LoveDA benchmarks compared with existing methods.**

***Index Terms*—Contrastive learning, distribution alignment, generative model, unsupervised domain adaptation (UDA) for semantic segmentation.**

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I. Introduction

SEMANTIC segmentation, which aims to partition remote sensing images into regions with explicit semantic information [1], has been widely applied to urban management, change detection, and environmental monitoring [2], [3], [4]. Despite the success of deep learning, its performance depends heavily on large-scale pixel-level annotations, whose collection is costly and time-consuming, especially in specialized fields [5].

Moreover, remote sensing data from various sensors, time periods, and regions exhibit complex intraclass variability [6], [7], which often gives rise to multipeaked intraclass feature distributions. This domain gap reduces the effectiveness of neural networks trained only on preannotated datasets when they are applied to real-world target environments [8]. To alleviate this issue, unsupervised domain adaptation (UDA) methods employ both existing annotated datasets and unlabeled data to facilitate adaptation to the target domain [9], [10]. However, a substantial performance gap to fully supervised training still remains.

Currently, most UDA methods for semantic segmentation follow the standard encoder–decoder paradigm, where pixel-wise features are extracted by an encoder and class predictions are produced by a dense discriminative classifier optimized using cross-entropy loss to estimate the posterior $p(c|f)$ [11], [12]. However, when annotations in the target domain are unavailable, relying solely on discriminative posterior estimation may be inadequate under severe domain shift.

In practice, this setting tends to expose two key limitations of purely discriminative UDA models. First, each category is typically represented by a single weight vector, which is insufficient to capture the pronounced intraclass diversity in remote sensing imagery. As shown in Fig. 1, substantial appearance variations within the same category often give rise to multiple intraclass subclusters in the feature space. Under a single-vector representation, these subclusters are easily mixed, and target-domain pixels with ambiguous boundaries are more likely to be aligned with incorrect subclusters.

Second, discriminative classifiers mainly learn decision boundaries for interclass separation, while cross-domain consistency among same-class subclusters is typically modeled only indirectly. Under UDA settings without target-domain annotations, same-class subclusters may therefore undergo

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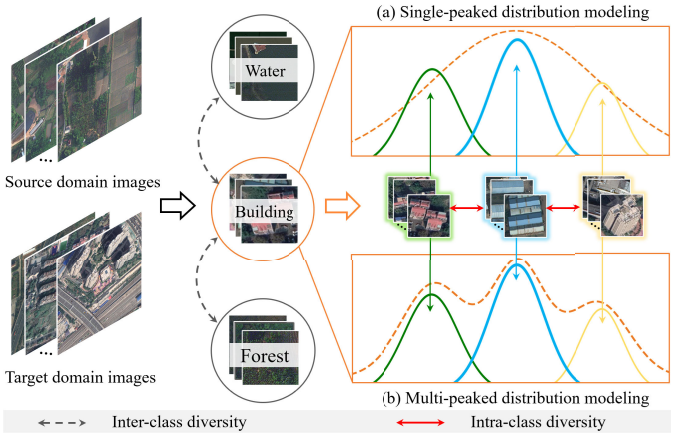


Fig. 1. Motivation for modeling multipeaked intraclass distributions in remote sensing. Pronounced intraclass variability gives rise to multiple subclusters within the same class, which are poorly captured by a single-vector representation in (a) but better modeled by multicomponent representations in (b).

center shifts and geometric mismatch across domains, which can aggravate cross-domain semantic inconsistency.

Motivated by these observations, we propose GMMDA, a Gaussian mixture model-based generative domain adaptation framework that explicitly models the class-conditional density $p(f|c)$ and predicts pixel labels through Bayesian inference. At the representation level, GMMDA uses multiple Gaussian components to characterize multipeaked intraclass feature structures in remote sensing imagery, with each component corresponding to an intraclass subcluster. This multicomponent formulation is intended to provide a finer grained description of intraclass structures and reduce subcluster mixing compared with a single-vector representation. As illustrated in Fig. 2, Gaussian mixture model (GMM)-based generative modeling offers a component-aware view for describing intraclass subcluster structures in the feature space.

Specifically, GMMDA first adopts the Sinkhorn EM [13] to learn class-wise GMMs from labeled source-domain features, thereby estimating $p(f|c)$. The posterior $p(c|f)$ is then derived via Bayes' rule and used for end-to-end training with cross-entropy supervision. Based on the learned Gaussian components, component-aware contrastive learning (CCL) treats them as fine-grained anchors and performs posterior-guided matching under a scale-aware distance metric, pulling target-domain features toward the most probable component while pushing them away from mismatched ones. To further encourage cross-domain consistency at the component level, we propose componentwise distribution alignment (CDA). For each class, CDA regularizes each source component against the corresponding source-guided target local statistics under a shared component index by jointly constraining their means and principal-subspace directions. In addition, a dynamic memory module (DMM) is incorporated to enrich source-domain statistics for GMM modeling, thereby providing more stable source components for CCL and CDA.

GMMDA explores a GMM-based generative classification framework for UDA in semantic segmentation and achieves competitive results on multiple benchmarks. More importantly, it suggests that class-conditional generative modeling can serve as a promising complement to discriminative UDA pipelines when intraclass structures are highly complex.

Our key contributions can be summarized as follows.

1) We propose a GMM-based generative UDA method for semantic segmentation that models the class-conditional density $p(f|c)$ via online Sinkhorn EM, predicts labels by Bayesian inference, and is trained end-to-end. Multiple Gaussian components are introduced for each class to better capture multipeaked intraclass variations and provide a finer grained alternative to single-vector representations.
2) CCL is introduced to treat Gaussian components as fine-grained semantic anchors to pull target-domain features toward the best-matched component while separating them from mismatched ones, improving intraclass separability.
3) We present CDA, which uses a shared component index to align source components with source-guided target local component statistics in both means and principal-subspace directions, thereby reducing cross-domain structural mismatch.

II. Related Work

A. Semantic Segmentation

Most semantic segmentation methods for remote sensing images are based on convolutional neural networks, which have shown strong capability for end-to-end training and pixel-wise classification since the introduction of fully convolutional networks by Long et al. [1]. The encoder–decoder architecture, first introduced by U-Net [14], has become one of the most widely adopted frameworks for semantic segmentation. To exploit multiscale image information and enlarge the receptive field, PSPNet [15] and DeepLab [16] introduce pyramid pooling and atrous convolution, respectively, thereby significantly improving segmentation performance. The integration of attention mechanisms further enables the adaptive fusion of local image features and global dependencies, addressing the limitations of convolutional kernels in modeling long-range relationships [17], [18].

Recently, transformer-based architectures have further advanced semantic segmentation and have shown strong capability in modeling long-range dependencies and multiscale context in both natural and remote sensing images [19], [20]. Qin et al. [21] propose a pyramid fusion transformer using multiscale features and a transformer-based decoder for improved mask-level classification for semantic segmentation. Li et al. [22] combine self-attention with a state-space model to capture both global dependencies and local structural details, thereby improving semantic segmentation in complex urban remote sensing scenes.

B. UDA for Semantic Segmentation

To reduce reliance on high-quality labels, considerable research has been devoted to UDA methods for semantic segmentation. Style transfer methods synthesize source-domain images with target-like appearance to reduce the visual

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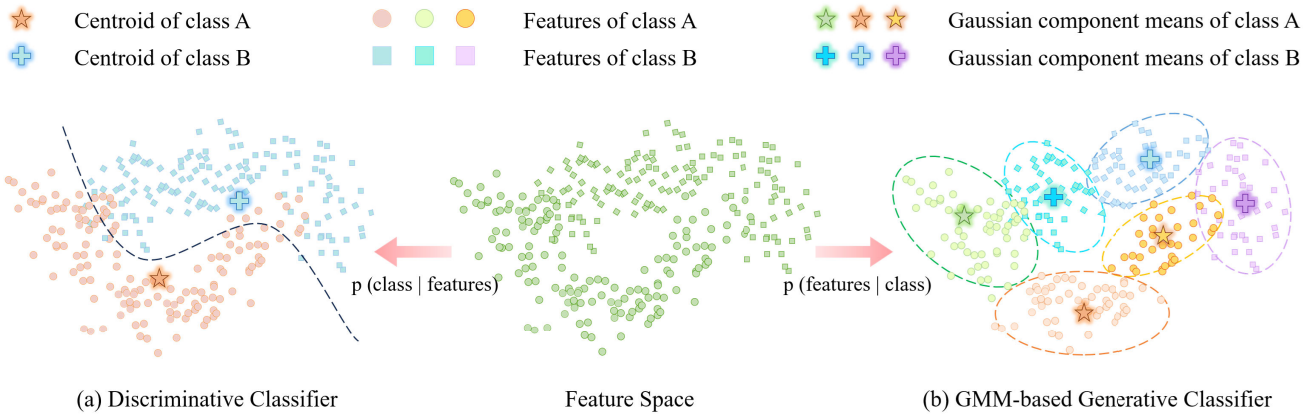


Fig. 2. Comparison between discriminative and generative modeling. (a) Discriminative modeling with a single-vector representation often underfits multipeaked intraclass distributions, making boundary-ambiguous pixels prone to misclassification. (b) Generative modeling with multiple Gaussian components can better characterize intraclass subclusters and yields component-level representations in the feature space, which may provide a finer grained basis for cross-domain alignment.

discrepancy between domains and improve target-domain generalization [23], [24]. Adversarial learning methods reduce the domain gap by aligning the feature-level or output-level distributions across domains [25], [26].

Self-training methods use pseudo-labels to facilitate target-domain learning [27], [28]. Representative works by Geng et al. [29], Hu et al. [30], and Hoyer et al. [31] further improve adaptation performance by combining pseudo-label-based self-training with stronger architectures and data perturbation strategies. Despite these differences, most UDA methods for semantic segmentation remain discriminative, predicting pixel labels via $p(c|f)$ and improving adaptation through cross-domain alignment or pseudo-label learning [32], [33]. As a result, their performance relies heavily on discriminative alignment under domain shift, while explicit modeling of class-conditional density remains relatively underexplored in this setting.

In contrast, our method models $p(f|c)$ with a generative classifier, which provides an explicit class-conditional description of class-wise feature distributions when intraclass structures are complex.

C. Contrastive Learning

Contrastive learning aims to learn discriminative representations by pulling positive pairs closer and pushing negative pairs farther apart. With the development of deep learning, contrastive learning has been increasingly applied to semantic segmentation. Wang et al. [34] propose a framework that learns from both annotated masks of base classes and pseudo masks of novel classes. Zhang et al. [35] propose a contrastive learning method with gradient-guided sampling to optimize learning and improve segmentation accuracy in remote sensing images. These methods impose contrastive constraints at the pixel-feature level, but their ability to model complex intraclass structures remains limited. To address this issue, some studies further adopt prototype-based contrastive learning, using class-level prototypes as semantic anchors to guide cross-domain alignment [36], [37]. Although such methods help improve class-level feature consistency, they are still limited by the coarse granularity of prototype representations and thus struggle to fully characterize the pronounced intraclass diversity in remote sensing scenes [38], [39].

In contrast, our method is built on class-conditional Gaussian mixture modeling and uses multiple Gaussian components within each class as fine-grained semantic references. This extends contrastive learning from the prototype level to the component level, thereby providing a more reliable reference for cross-domain alignment.

D. GMMs in UDA

GMMs have been introduced into UDA to better characterize complex feature distributions and alleviate the limitations of single-center representations [40], [41]. Existing studies typically use GMMs in the feature space to model intraclass variations or construct more informative prototypes for cross-domain alignment [42], [43]. However, these methods usually use GMMs as auxiliary tools for feature modeling, clustering, or alignment, while dense prediction is still performed by a discriminative classifier.

Different from these works, GMMDA uses class-conditional GMMs as generative dense classifiers and computes pixel-wise Bayesian posteriors for prediction. The learned Gaussian components are further used as semantic units for CCL and CDA, extending their role from feature modeling to dense prediction and component-level adaptation.

III. Methodology

A. Overview

In UDA, we define the source-domain images as $X_S = \{x_S^i\}_{i=1}^{N_S}$ with annotations $Y_S = \{y_S^i\}_{i=1}^{N_S}$, and the unlabeled target-domain images as $X_T = \{x_T^i\}_{i=1}^{N_T}$. Let C denote the number of classes, where $c \in \{1, \ldots, C\}$. Our goal is to learn a high-performance segmentation model that generalizes to the target domain without requiring target-domain annotations.

Fig. 3 illustrates the proposed method. We learn class-wise GMMs from source-domain features through an online momentum Sinkhorn EM procedure, where each class is

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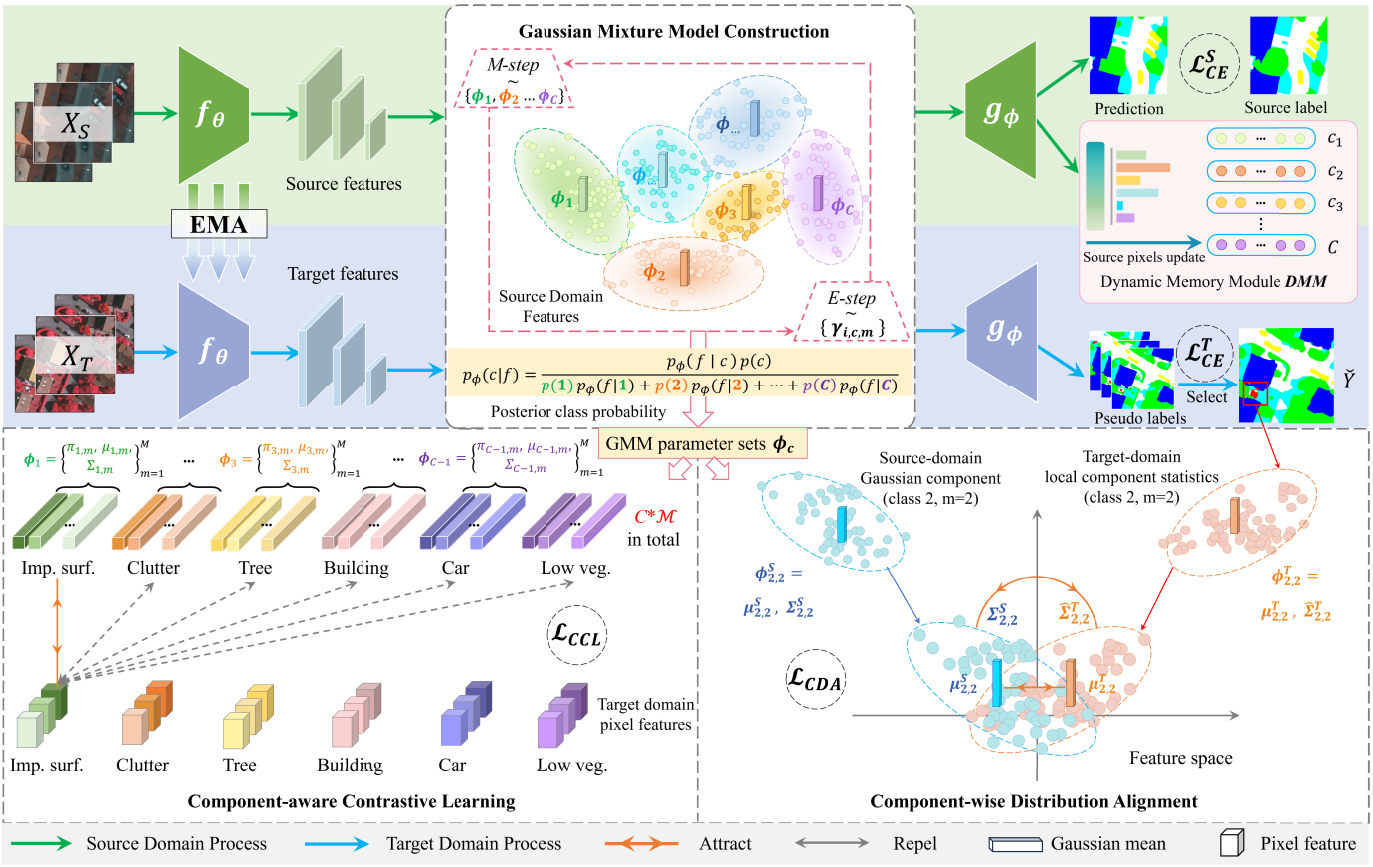


Fig. 3. Overview of GMMDA. Class-conditional GMMs are learned from source features via online Sinkhorn EM, and pixel-wise posteriors $p(c|f)$ are computed by Bayesian inference for end-to-end training. Using the learned components as fine-grained semantic references, CCL assigns each target feature to its most probable component based on responsibilities and pushes it away from mismatched components. CDA further aligns source components with source-guided target local statistics under a shared component index by jointly constraining their means and principal-subspace directions. A DMM stores representative source-domain features to improve the stability of GMM learning, thereby providing more reliable source components for CCL and CDA.

modeled by M Gaussian components to capture multipeaked intraclass feature distributions. Based on the learned class-conditional densities, the GMM-based generative classifier performs Bayesian inference to obtain pixel-wise posteriors $p(c|f)$ for both domains. We update ϕ via EM and optimize θ end-to-end by back-propagation under $\mathcal{L}_{\text{CE}}^{S} + \mathcal{L}_{\text{CE}}^{T}$ (Section III-B). Moreover, we exploit the learned Gaussian components to perform component-level matching for target-domain features. CCL encourages target-domain features to align with their best-matched component while remaining separable from other intraclass subclusters (Section III-C). Furthermore, CDA promotes structure-aware consistency by jointly constraining source components and corresponding target local statistics in terms of their means and principal-subspace directions (Section III-D).

B. GMM Construction

Most existing UDA methods focus on cross-domain feature alignment while retaining a discriminative dense classifier for pixel prediction. In contrast, we replace the conventional discriminative classifier with a GMM-based generative classifier that explicitly models the class-conditional density $p_\phi(f|c)$. This formulation allows pixel-wise posterior $p_\phi(c|f)$ to be computed through Bayes' rule and provides a distribution-aware basis for subsequent component-level alignment.

In the following, $p_\phi(\cdot)$ denotes a GMM-parameterized distribution, where ϕ collects the mixture parameters. Specifically, we use class prior $p(c)$ and the class-conditional density $p(f|c)$ to model the joint distribution $p(f, c)$

$$p(f, c) = p(f|c)\, p(c). \tag{1}$$

The class prior reflects the overall prevalence of each class and plays an important role in Bayesian inference for distribution modeling. In remote sensing imagery, severe long-tailed class distributions can bias training toward frequent classes and degrade the performance of rare categories. To alleviate this problem, we adopt rare class sampling (RCS) [33] during source-domain mini-batch construction. RCS increases the sampling probability of source images containing rare classes, thereby increasing the occurrence frequency of under-represented categories in source-domain mini-batches. It is a sampling-level rebalancing strategy rather than a modification to the mathematical form of posterior inference or an additional class-weighted loss. The posterior probability of class c is computed as

$$p_\phi(c \mid f) = \frac{p_\phi(f \mid c)\; p(c)}{\sum_{c'=1}^{C} p(c')\; p_\phi(f \mid c')}. \tag{2}$$

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To capture multipeaked intraclass feature distributions, we model the class-conditional density of each class as an M-component Gaussian mixture

$$p(f|c;\phi_c) = \sum_{m=1}^{M} \pi_{c,m} \mathcal{N}\left(f;\mu_{c,m},\Sigma_{c,m}\right). \tag{3}$$

Here, $\pi_{c,m}$ denotes the component prior of the mth Gaussian component in class c. Following the balanced component assumption in Sinkhorn EM, we fix $\pi_{c,m} = 1/M$ for all components and do not update it during EM. The parameters $\mu_{c,m}$ and $\Sigma_{c,m}$ denote the corresponding mean and covariance, respectively. The GMM is parameterized by $\phi = \{\phi_c\}_{c=1}^{C}$, where $\phi_c = \{\mu_{c,m},\Sigma_{c,m}\}_{m=1}^{M}$. By assigning an individual mean and covariance to each component, the model can represent diverse intraclass characteristics more flexibly.

However, standard full-dataset EM is computationally intractable for dense semantic segmentation, because each mini-batch contains a large number of pixel-level features and the feature space continuously evolves during training. Therefore, we adopt an online momentum Sinkhorn EM strategy, where the GMM parameters are updated once at each training iteration by performing only one Sinkhorn-normalized EM step instead of running EM to convergence.

For each class c, let $\mathcal{B}_c^S$ denote the source-domain pixel features of class c in the current mini-batch, and let $\mathcal{M}_{c,m}^S$ denote the detached source-domain features stored in the DMM for the mth Gaussian component. The feature set used for the current online GMM update is defined as

$$\mathcal{Q}_c^S = \mathcal{B}_c^S \cup \left(\bigcup_{m=1}^{M} \mathcal{M}_{c,m}^S\right). \tag{4}$$

Given $\mathcal{Q}_c^S$, we update the class-wise GMM parameters through the following online Sinkhorn EM procedure.

E-Step: For each feature $f_i \in \mathcal{Q}_c^S$, we compute its score with respect to the mth Gaussian component as

$$s_{i,c,m} = \log \pi_{c,m} + \log \mathcal{N}\left(f_i;\mu_{c,m}^{(t-1)},\Sigma_{c,m}^{(t-1)}\right). \tag{5}$$

Since the component prior is fixed as $\pi_{c,m} = 1/M$, the score is mainly determined by the component likelihood. Instead of directly applying a softmax over the component scores, we use Sinkhorn normalization to obtain balanced component assignment weights

$$R_c = \text{Sinkhorn}\left(\exp\left(S_c\right)\right) \tag{6}$$

where $S_c = [s_{i,c,m}]$ and $R_c = [r_{i,c,m}^S]$. The resulting assignment matrix satisfies

$$\sum_{m=1}^{M} r_{i,c,m}^S = 1, \quad \sum_{f_i \in \mathcal{Q}_c^S} r_{i,c,m}^S \approx \frac{\left|\mathcal{Q}_c^S\right|}{M}. \tag{7}$$

This balanced assignment constraint reduces the risk of degenerate assignments, where most features are assigned to only a few dominant components.

M-Step: Given the assignment weights, the effective sample size of the mth component in class c is computed as

$$N_{c,m} = \sum_{f_i \in \mathcal{Q}_c^S} r_{i,c,m}^S. \tag{8}$$

We then estimate the temporary mean and covariance of each component as

$$\hat{\mu}_{c,m} = \frac{1}{N_{c,m}} \sum_{f_i \in \mathcal{Q}_c^S} r_{i,c,m}^S f_i \tag{9}$$

$$\hat{\Sigma}_{c,m} = \frac{1}{N_{c,m}} \sum_{f_i \in \mathcal{Q}_c^S} r_{i,c,m}^S \left(f_i - \hat{\mu}_{c,m}\right)\left(f_i - \hat{\mu}_{c,m}\right)^{\top} + \epsilon I \tag{10}$$

where ϵI is added for numerical stability.

To stabilize GMM estimation in the continuously evolving feature space, we update the GMM parameters by momentum smoothing

$$\mu_{c,m}^{(t)} = \rho \mu_{c,m}^{(t-1)} + (1-\rho)\hat{\mu}_{c,m} \tag{11}$$

$$\Sigma_{c,m}^{(t)} = \rho \Sigma_{c,m}^{(t-1)} + (1-\rho)\hat{\Sigma}_{c,m} \tag{12}$$

where ρ denotes the momentum coefficient and is set to 0.999 in all experiments. This online update avoids repeated full-dataset re-estimation and eliminates the need to run EM to convergence at each iteration.

During training, we use a class-wise GMM as a generative classifier to compute pixel-wise posteriors. Given a source image $x_S^{(i)}$, the encoder f_θ extracts pixel features $f^{(i,j)} = f_\theta(x_S^{(i)})_j$, where $j \in \{1,\ldots,HW\}$, and the GMM classifier outputs $p_\phi(c \mid f^{(i,j)})$. Let $y_S^{(i,j)} \in \{1,\ldots,C\}$ denote the ground-truth label of the jth pixel in the ith source image. The source-domain cross-entropy loss is defined as

$$\mathcal{L}_{\text{CE}}^S = -\frac{1}{N_S HW} \sum_{i=1}^{N_S} \sum_{j=1}^{HW} \log p_\phi\left(y_S^{(i,j)} \middle| f_\theta\left(x_S^{(i)}\right)_j\right). \tag{13}$$

For each target image, we define the prediction confidence and pseudo-label as

$$p_T^{(i,j)} = \max_c \; p_\phi\left(c \middle| f_\theta\left(x_T^{(i)}\right)_j\right)$$

$$\hat{c}_T^{(i,j)} = \arg\max_c \; p_\phi\left(c \middle| f_\theta\left(x_T^{(i)}\right)_j\right). \tag{14}$$

Reliable pseudo-labels are then selected using a confidence mask

$$m_T^{(i,j)} = \mathbf{1}\left[p_T^{(i,j)} \geq t\right] \tag{15}$$

where t is the confidence threshold. The target-domain self-training loss is computed over selected pixels

$$\mathcal{L}_{\text{CE}}^T = -\frac{1}{\sum_{i=1}^{N_T}\sum_{j=1}^{HW} m_T^{(i,j)}} \sum_{i=1}^{N_T}\sum_{j=1}^{HW} m_T^{(i,j)} \cdot \log p_\phi\left(\hat{c}_T^{(i,j)} \middle| f_\theta\left(x_T^{(i)}\right)_j\right). \tag{16}$$

The GMM parameters ϕ are updated by the Sinkhorn EM, while the encoder parameters θ are optimized through back-propagation under $\mathcal{L}_{\text{CE}}^S + \mathcal{L}_{\text{CE}}^T$.

To obtain stable statistics for online GMM estimation, we introduce a DMM to store representative source-domain pixel features from previous mini-batches. The DMM is updated using source-domain features only, since target-domain pseudo-labels may be noisy.

The DMM is organized at the class-component level. For each class c and Gaussian component m, an independent first-in, first-out (FIFO) queue $\mathcal{M}_{c,m}^S$ is maintained with a maximum

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capacity of 32K pixel features per queue. This design preserves component-specific historical samples and reduces the mixing of different intraclass modes. The stored features are detached and used only for GMM estimation. Therefore, the DMM does not participate in back-propagation and is discarded during inference.

At each training iteration, we retain source pixels whose ground-truth labels are consistent with the current predictions. These features are assigned to Gaussian components according to the current GMM-based component assignment and are then inserted into the corresponding class-component queues. If the number of candidates exceeds the sampling budget, sparse uniform sampling is performed before insertion. When a queue exceeds its capacity, the oldest features are removed in FIFO order. Sparse sampling reduces the redundancy of dense remote sensing segmentation features, since neighboring pixels in large homogeneous regions often have similar representations. In the default setting, at most 100 features are sampled for each class at each memory update.

C. Component-Aware Contrastive Learning

In CCL, responsibilities derived from the source-domain class-conditional GMM guide component assignment for target-domain pixels, avoiding the coarse-grained alignment induced by a single-vector representation. The most probable component is used as the positive anchor, while the others serve as negatives in the contrastive loss, promoting more appropriate intraclass associations under domain shift.

As illustrated in Fig. 4, the source-domain GMM for class c is represented by

$$\phi_c^S = \left\{\left(\mu_{c,m}^S, \Sigma_{c,m}^S\right)\right\}_{m=1}^{M} \tag{17}$$

where $\mu_{c,m}^S \in \mathbb{R}^d$ denotes the mean of the mth Gaussian component, and $\Sigma_{c,m}^S$ is the corresponding covariance matrix. Under the balanced component assumption, the component prior is fixed as $\pi_{c,m}^S = 1/M$. To balance representational capacity and computational efficiency, we parameterize the covariance of the mth Gaussian component in class c using a low-rank-plus-diagonal form

$$\Sigma_{c,m}^S = U_{c,m}\,\Lambda_{c,m}\,U_{c,m}^\top + D_{c,m} \tag{18}$$

where $U_{c,m} \in \mathbb{R}^{d\times r}$ spans a low-dimensional principal-variation subspace of the mth component, $\Lambda_{c,m}$ controls the subspace scales, and $D_{c,m}$ is a diagonal residual term for numerical stability.

For notational simplicity, we use a single index i to denote each selected target-domain feature. We then set $c = \hat{c}_T^{(i)}$ for f_i using its pseudo-label in (14), and compute its component responsibility under the source-domain GMM as

$$\gamma_{i,c,m}^T = p\left(m \mid f_i, c; \phi_c^S\right) = \frac{\mathcal{N}\left(f_i; \mu_{c,m}^S, \Sigma_{c,m}^S\right)}{\sum_{j=1}^{M} \mathcal{N}\left(f_i; \mu_{c,j}^S, \Sigma_{c,j}^S\right)}. \tag{19}$$

This responsibility quantifies the association of f_i with each component in class c and is used to select the most probable component for subsequent alignment. Accordingly, we select the component index via the maximum a posteriori criterion

$$m_i^* = \arg\max_m \gamma_{i,c,m}^T \tag{20}$$

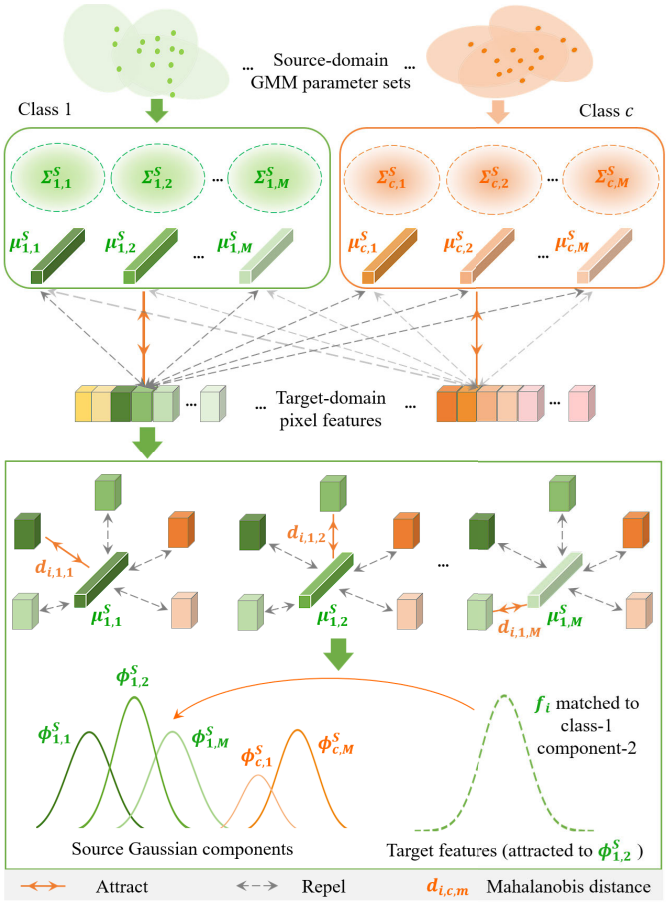


Fig. 4. Illustration of CCL. Source-domain class-wise GMM components serve as fine-grained semantic references for computing matching scores for target-domain features. CCL selects the best-matched Gaussian component as the positive anchor and pulls each target feature toward that component (e.g., $f_i \rightarrow \phi_{1,2}^S$). Here, the component covariance shapes the similarity metric, enabling more reliable component-aware matching than mean-only comparison.

and treat the assigned component $\phi_{c,m_i^*}^S = (\mu_{c,m_i^*}^S, \Sigma_{c,m_i^*}^S)$ as the positive anchor for pixel-level alignment.

Next, we construct a component-aware contrastive objective that pulls target features toward their matched positive components while pushing them away from mismatched components. The component-aware contrastive loss is defined as

$$\mathcal{L}_{\mathrm{CCL}} = -\frac{1}{\tau}\,\mathrm{sim}\left(f_i, \phi_{c,m_i^*}^S\right) + \log\left(\exp\left(\mathrm{sim}\left(f_i, \phi_{c,m_i^*}^S\right)/\tau\right) + \sum_{\phi^-\in\mathcal{P}_i^-} \exp\left(\mathrm{sim}\left(f_i, \phi^-\right)/\tau\right)\right) \tag{21}$$

where τ is a temperature hyperparameter, and $\mathcal{P}_i^-$ denotes the negative set consisting of Gaussian components $\{\phi_{c,m}^S\}$ from 1) the same class but different component indices, $\{\phi_{c,m}^S\}_{m\neq m_i^*}$ and 2) all other classes, $\{\phi_{c',m}^S\}_{c'\neq c}$.

Unlike approaches that use only a single class centroid as the contrastive target, the responsibility-guided adaptive component assignment avoids forcing distinct intraclass

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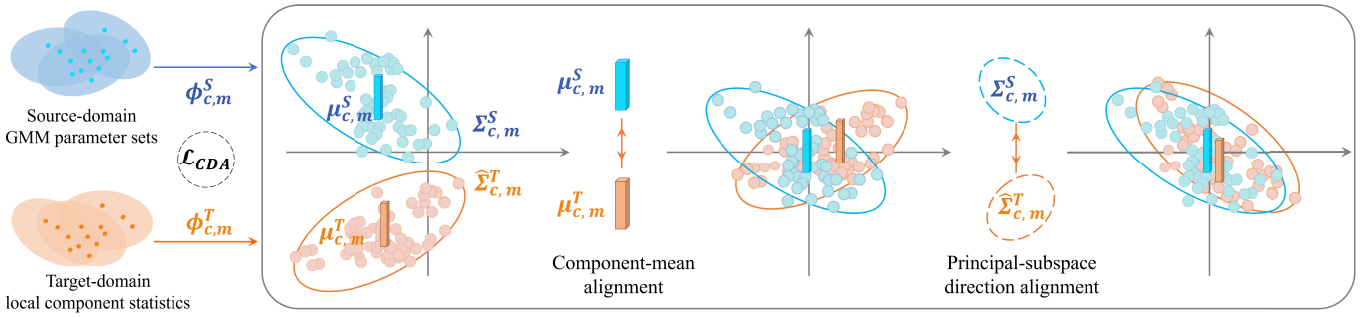


Fig. 5. Illustration of CDA. CDA jointly constrains the means and principal-subspace directions of source-domain component $\phi^S_{c,m}$ and the corresponding source-guided target local statistics $\phi^T_{c,m}$ under a shared component index. In this way, alignment is extended from centroid attraction to distribution-structure calibration, which helps mitigate within-class structural mismatch.

subclusters to collapse into a single center, thereby better preserving intraclass subcluster structure and distribution separability.

To make the alignment reflect not only mean proximity but also intracomponent scale and principal directions, we measure similarity using the Mahalanobis distance

$$d_{i,c,m} = \left(f_i - \mu^S_{c,m}\right)^\top \left(\Sigma^S_{c,m}\right)^{-1} \left(f_i - \mu^S_{c,m}\right)$$
$$\text{sim}\left(f_i, \phi^S_{c,m}\right) = -d_{i,c,m}. \tag{22}$$

Compared with Euclidean similarity, the Mahalanobis metric downweights directions with high uncertainty and imposes stronger penalties along more discriminative directions with low variance. This mitigates the systematic bias introduced by mean-only alignment and further improves alignment stability and robustness.

D. Componentwise Distribution Alignment

Since target-domain component identities are unavailable, CDA, as illustrated in Fig. 5, estimates local target-component statistics from pseudo-label confidence and source-guided responsibilities, and uses a shared component index to regularize them against source components through joint constraints on means and principal subspaces, thereby alleviating structural mismatch.

Specifically, for each class $c \in \{1, \ldots, C\}$, we obtain source-domain class-wise GMM parameter set $\{(\mu^S_{c,m}, \Sigma^S_{c,m})\}_{m=1}^M$. To balance expressive power and computational tractability, we adopt the low-rank-plus-diagonal covariance parameterization in (18) and use the corresponding principal subspace $U^S_{c,m}$ to characterize the geometry of the mth component. For each mini-batch, given the set of pseudo-labeled target-domain samples belonging to class c, denoted by $\tilde{\Omega}^T_c$, we compute weighted estimates of the local target-component statistics using the prediction confidence $p^{(i)}_T$ and posterior responsibility $\gamma^T_{i,c,m}$ defined in (19), thereby obtaining a more robust characterization of the local distribution. We define the weighting coefficient as $a_{i,c,m} = p^{(i)}_T \gamma^T_{i,c,m}$. The target-domain mean and covariance of the mth component are then estimated as

$$\mu^T_{c,m} = \frac{\sum_{i\in\tilde{\Omega}^T_c} a_{i,c,m} f_i}{\sum_{i\in\tilde{\Omega}^T_c} a_{i,c,m}}$$
$$\hat{\Sigma}^T_{c,m} = \frac{\sum_{i\in\tilde{\Omega}^T_c} a_{i,c,m} \left(f_i - \mu^T_{c,m}\right)\left(f_i - \mu^T_{c,m}\right)^\top}{\sum_{i\in\tilde{\Omega}^T_c} a_{i,c,m}} + \varepsilon I. \tag{23}$$

Here, εI is added for numerical stability, with $\varepsilon = 10^{-5}$ in all experiments. We then perform eigen-decomposition on $\hat{\Sigma}^T_{c,m}$ to obtain the principal-subspace basis $U^T_{c,m}$, and represent the subspace direction using the projection matrices

$$P^S_{c,m} = U^S_{c,m}\left(U^S_{c,m}\right)^\top, \quad P^T_{c,m} = U^T_{c,m}\left(U^T_{c,m}\right)^\top. \tag{24}$$

Here, $U^S_{c,m}$ and $U^T_{c,m}$ are constructed from the top-r eigenvectors of the corresponding covariance matrices and are thus column-orthonormal.

Considering pseudo-label noise in the target domain and the uncertainty of local component estimation, we further introduce a responsibility-based weight to adaptively modulate the contribution of each component to the alignment

$$\omega_{c,m} = \frac{1}{\left|\tilde{\Omega}^T_c\right|} \sum_{i\in\tilde{\Omega}^T_c} \gamma^T_{i,c,m}. \tag{25}$$

On this basis, we impose both mean consistency and principal-subspace consistency between each source component and the target local statistics estimated under the shared index (c, m)

$$\mathcal{L}_{\text{CDA}} = \sum_{c=1}^{C}\sum_{m=1}^{M} \omega_{c,m}\left(\left\|\mu^S_{c,m} - \mu^T_{c,m}\right\|_2^2 + \left\|P^S_{c,m} - P^T_{c,m}\right\|_F^2\right). \tag{26}$$

CDA mitigates error accumulation caused by incorrect target pseudo-labels through source-guided and reliability-weighted local statistic estimation. Target pseudo-labels are not used to update the source-domain GMM components, which are estimated from labeled source features and serve as stable semantic references. Moreover, CDA estimates target local statistics only from high-confidence target-domain pixels, weights selected features by $a_{i,c,m}$, and further modulates component-level alignment by $\omega_{c,m}$. Therefore, uncertain predictions or weakly matched target features have limited influence on the CDA objective, reducing the risk of error accumulation during early adaptation.

E. Overall Loss Function

In summary, the overall loss function of the proposed method is formulated as follows:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}^S_{\text{CE}} + \mathcal{L}^T_{\text{CE}} + \alpha\mathcal{L}_{\text{CCL}} + \beta\mathcal{L}_{\text{CDA}}. \tag{27}$$

Here, α and β balance the contributions of CCL and CDA, respectively. Their effects are analyzed in Section IV-D.

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TABLE I
RESULTS OF DIFFERENT UDA METHODS ON THE POTSDAM ⟷ VAIHINGEN TASK

Task	Method	Back./Arch.	Imp. surf.	Clutter	Tree	Building	Car	Low veg.	OA (%)	mIoU (%)
Potsdam ↓ Vaihingen	Source Only [NeurIPS21][44]	MiT-B5	40.12	74.08	25.91	67.03	16.24	11.94	59.72	39.22
	Oracle [NeurIPS21][44]	MiT-B5	79.23	88.27	65.06	76.70	70.84	60.30	83.44	73.40
	AdaptSegNet [CVPR18][45]	ResNet-101	63.89	70.12	32.54	59.31	9.72	6.00	67.36	40.26
	CLAN [CVPR19][25]	ResNet-101	58.15	70.70	38.76	62.06	9.42	6.79	69.81	40.98
	ABFNet [TETCI26][46]	Res2Net-101	62.57	24.28	**54.67**	**75.77**	27.84	32.87	72.26	46.33
	JDAF [TGRS24][47]	MiT-B5	76.45	84.49	39.19	65.09	60.35	39.26	73.82	60.81
	ClA-UDA [TGRS24][48]	MiT-B5	77.25	85.89	51.07	69.73	60.08	28.64	81.53	62.11
	PFM-JONet [TGRS25][49]	MiT-B5 + SAM	77.13	85.56	45.89	67.92	60.09	39.59	81.42	62.70
	AACDN [JSTAR26][50]	MiT-B5	75.68	85.69	44.11	66.99	57.31	22.32	80.61	58.68
	SSC-CycleGAN [JSTAR26][51]	MiT-B5	74.43	85.30	36.13	63.57	47.89	13.55	72.36	53.48
	Hu et al. [JSTAR26][30]	MiT-B5	76.74	84.75	47.00	70.44	59.70	37.33	80.73	62.66
	Ours	MiT-B5	**77.50**	**86.39**	45.76	66.06	**60.39**	**49.84**	**82.38**	**64.32**
Vaihingen ↓ Potsdam	Source Only [NeurIPS21][44]	MiT-B5	50.10	74.94	21.74	38.39	41.56	6.76	51.05	38.92
	Oracle [NeurIPS21][44]	MiT-B5	84.01	91.01	74.30	73.07	82.40	27.32	79.88	72.02
	AdaptSegNet [CVPR18][45]	ResNet-101	58.32	60.31	52.77	23.61	38.53	1.82	64.07	39.23
	CLAN [CVPR19][25]	ResNet-101	65.62	65.98	57.09	33.92	37.70	1.66	67.08	43.66
	ABFNet [TETCI26][46]	Res2Net-101	60.75	1.17	36.19	**75.96**	45.94	**48.55**	70.84	44.76
	JDAF [TGRS24][47]	MiT-B5	**77.25**	81.10	63.31	58.15	71.65	10.05	81.90	60.25
	ClA-UDA [TGRS24][48]	MiT-B5	74.08	86.11	63.33	58.02	72.91	10.54	83.34	60.83
	PFM-JONet [TGRS25][49]	MiT-B5 + SAM	75.80	87.05	63.60	56.29	68.94	6.43	83.81	59.69
	AACDN [JSTAR26][50]	MiT-B5	72.39	81.57	63.07	46.39	72.58	2.97	81.15	56.49
	SSC-CycleGAN [JSTAR26][51]	MiT-B5	64.20	76.47	58.64	52.15	61.35	9.60	77.87	53.73
	Hu et al. [JSTAR26][30]	MiT-B5	75.40	85.96	63.39	57.41	71.73	11.03	84.01	60.82
	Ours	MiT-B5	76.60	**88.87**	**64.44**	59.52	**73.44**	10.38	**85.63**	**62.21**

IV. EXPERIMENTS AND RESULTS

In this section, we first introduce the datasets and evaluation metrics used in the experiments. We then compare our method with state-of-the-art approaches, followed by comprehensive ablation studies, parameter analysis, and feature visualization of the proposed network.

A. Datasets and Evaluation Metric

1) Datasets: We conduct domain adaptation experiments in four distinct configurations across two datasets.

1) *ISPRS 2-D:* The international society for photogrammetry and remote sensing (ISPRS) 2-D dataset [52], provided by the International Society for Photogrammetry and Remote Sensing, includes the Potsdam and Vaihingen subsets, both annotated with six semantic categories. Potsdam contains 38 aerial images at 5 cm resolution, each of size 6000 × 6000 pixels, whereas Vaihingen contains 33 images at 9 cm resolution, each of approximately 2000×2000 pixels. This dataset is widely used in remote sensing semantic segmentation and is also commonly adopted in UDA studies to evaluate model adaptability across different urban scenes.
2) *LoveDA Dataset:* The LoveDA dataset [53] consists of 5987 high-resolution optical remote sensing images from Nanjing, Changzhou, and Wuhan, China, and is designed for remote sensing semantic segmentation. Each image has a spatial resolution of 0.3 m per pixel and dimensions of 1024×1024 pixels. The dataset covers both urban and rural landscapes across seven categories, testing model adaptability and generalization under complex backgrounds and multiscale objects. A notable characteristic of LoveDA is its severe class imbalance, which poses an additional challenge for UDA.

2) Evaluation Metrics: To evaluate the proposed method, we use overall pixel accuracy (OA) and mean intersection over union (mIoU). OA measures accuracy over all pixels and categories. mIoU is defined as the intersection between the prediction and the ground truth divided by their union, and serves as the primary criterion for evaluating each method.

B. Comparisons With State-of-the-Art Methods

In this study, we evaluate our method under four dataset configurations and quantitatively compare it with classical UDA baselines and recent remote sensing UDA approaches, as reported in Tables I and II. To improve transparency, we explicitly report the backbone or main architecture used by each compared method in the tables. The compared methods cover representative directions, including adversarial adaptation, input-space adaptation, foundation-model-based adaptation, and multimodal adaptation. Most recent remote sensing UDA methods in our comparison use MiT-B5 or SegFormer-based architectures, while several methods follow different architectures according to their original designs, such as ResNet-101 for AdaptSegNet and category-level adversarial network (CLAN), MiT-B5 + segment anything model (SAM) for PFM-JONet, and Res2Net-101 with multimodal inputs for ABFNet. Therefore, the quantitative results should be interpreted as method-level comparisons with explicit backbone or architecture information rather than strictly backbone-controlled comparisons, since differences in backbone or architecture may still affect absolute performance. For qualitative analysis, we present visual results

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TABLE II
RESULTS OF DIFFERENT UDA METHODS ON THE RURAL ⟷ URBAN TASK. "–" INDICATES THAT THE METHOD IS NOT APPLICABLE BECAUSE THE REQUIRED DSM/DEPTH MODALITY IS UNAVAILABLE

Task	Method	Back./Arch.	Background	Building	Road	Water	Barren	Forest	Agr.	OA (%)	mIoU (%)
Rural ↓ Urban	Source Only [NeurIPS21][44]	MiT-B5	39.82	46.21	33.61	67.12	25.73	19.01	23.65	42.08	36.45
	Oracle [NeurIPS21][44]	MiT-B5	49.21	65.07	53.02	69.59	26.04	37.86	45.97	62.41	49.54
	AdaptSegNet [CVPR18][45]	ResNet-101	37.53	55.05	39.77	65.02	21.91	20.53	19.73	40.41	37.08
	CLAN [CVPR19][25]	ResNet-101	40.91	44.92	34.67	67.10	23.95	21.43	53.62	59.41	40.94
	ABFNet [TETCI26][46]	Res2Net-101	-	-	-	-	-	-	-	-	-
	JDAF [TGRS24][47]	MiT-B5	37.81	53.12	50.33	64.27	39.64	42.98	54.41	64.77	48.94
	ClA-UDA [TGRS24][48]	MiT-B5	**41.96**	57.77	61.48	72.04	39.54	44.69	53.61	67.14	53.01
	PFM-JONet [TGRS25][49]	MiT-B5 + SAM	41.44	**59.28**	**64.03**	72.72	38.89	43.44	55.46	67.30	53.61
	AACDN [JSTAR26][50]	MiT-B5	37.35	52.95	59.57	72.55	37.15	42.61	60.00	66.78	51.74
	SSC-CycleGAN [JSTAR26][51]	MiT-B5	36.50	47.29	54.74	70.85	38.32	44.38	54.08	64.42	49.45
	Hu et al. [JSTAR26][30]	MiT-B5	39.29	58.29	60.79	73.03	36.34	39.54	56.93	66.93	52.03
	Ours	MiT-B5	39.96	54.24	58.34	**73.09**	**39.90**	**45.38**	**67.66**	**68.45**	**54.08**
Urban ↓ Rural	Source Only [NeurIPS21][44]	MiT-B5	37.89	24.91	26.10	42.63	12.89	17.02	28.41	36.97	27.12
	Oracle [NeurIPS21][44]	MiT-B5	56.46	57.63	40.87	63.59	28.92	26.84	65.24	59.01	48.51
	AdaptSegNet [CVPR18][45]	ResNet-101	40.36	22.81	31.74	59.11	**29.06**	14.42	24.38	38.02	31.70
	CLAN [CVPR19][25]	ResNet-101	41.83	37.21	34.77	62.42	18.96	19.48	29.55	42.37	34.89
	ABFNet [TETCI26][46]	Res2Net-101	-	-	-	-	-	-	-	-	-
	JDAF [TGRS24][47]	MiT-B5	54.77	43.05	39.64	67.52	23.41	35.12	51.83	64.94	45.05
	ClA-UDA [TGRS24][48]	MiT-B5	51.23	47.03	46.47	70.72	13.35	37.57	35.16	64.13	43.08
	PFM-JONet [TGRS25][49]	MiT-B5 + SAM	47.79	**57.72**	**46.78**	72.12	14.55	31.17	37.71	62.67	43.98
	AACDN [JSTAR26][50]	MiT-B5	45.47	42.28	42.81	71.84	6.62	33.22	51.66	65.37	41.98
	SSC-CycleGAN [JSTAR26][51]	MiT-B5	33.48	46.59	40.90	65.75	9.83	24.80	45.24	56.54	38.08
	Hu et al. [JSTAR26][30]	MiT-B5	52.31	54.50	42.41	70.06	10.13	36.45	51.94	69.76	45.40
	Ours	MiT-B5	**55.54**	50.55	40.18	**72.85**	14.18	**38.25**	**52.78**	**71.79**	**46.33**

of our method alongside selected representative approaches in Figs. 6–9.

1) *Potsdam ⟶ Vaihingen Task:* Across different urban environments, most classes exhibit substantial variations in color, shape, and texture. In addition, certain classes, such as "Low vegetation" ("Low veg.") and "Tree," frequently co-occur and share similar visual characteristics, making accurate boundary delineation particularly challenging. Table I (top) presents a comparison between our method and state-of-the-art approaches on the Potsdam→Vaihingen task. The results show that, compared to methods that use only the source domain, our approach improves the overall metrics and yields gains on several individual classes. Our model achieves an mIoU of 64.32%, representing an improvement of 25.1% over the source-only method and a 1.62% gain over the previous best method. Notably, for the challenging class "Low veg," our method surpasses the other approaches by a considerable margin, achieving an improvement of 10.25%.
Fig. 6 provides a visual comparison of segmentation results produced by different models. Our model better distinguishes intraclass visual variations and reduces confusion between categories with similar appearances. For instance, in the first row of Fig. 6, our method better delineates the boundary between "Clutter" (red) and "Tree" (green) while correctly preserving the "Car" (yellow) region. In rows two to four, GMMDA clearly separates "Tree," "Building," and "Low veg," whereas other methods exhibit different degrees of confusion.

2) *Vaihingen ⟶ Potsdam Task:* Table I (bottom) reports the comparative results on the Vaihingen→Potsdam task. In this reverse adaptation setting, our method also achieves the best overall metrics and remains competitive in most categories. Fig. 7 provides a visual comparison of segmentation results from different models. In the first row, our method better delineates the boundary of the "Car" class within the "Low veg." area. In the second row, our method shows less confusion between "Tree" and "Low veg," while other methods produce more ambiguous predictions in the corresponding regions. Similarly, our method maintains smoother and more coherent boundaries for the class "Building" (dark blue) in the third and fourth rows.

3) *LoveDA Rural ⟶ LoveDA Urban Task:* To verify the effectiveness of our method in diverse and challenging remote-sensing environments, we further conduct experiments on the LoveDA dataset. The rural and urban subsets were collected from multiple regions in China, including Nanjing, Changzhou, and Wuhan, resulting in complex multipeaked representations among similar instances. This challenge is further compounded by the severe category imbalance between the rural and urban domains. For example, the "Agriculture" and "Forest" samples are 17 times and six times more prevalent in the rural subset than in the urban subset, respectively, while the proportion of "Building" samples in rural is only one-seventh that in urban.
The results for the Rural→Urban task are shown in Table II (top), where the rural dataset mainly consists of

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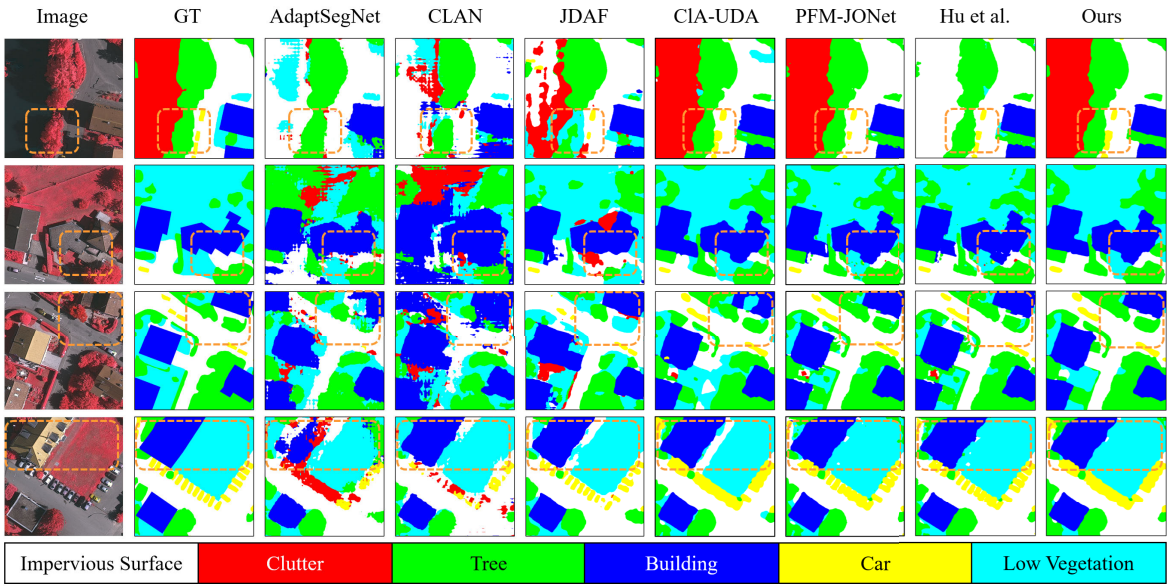


Fig. 6. Visual comparison on the Potsdam → Vaihingen task. From left to right: input images, ground-truth segmentation maps, results of state-of-the-art methods, and results of our method.

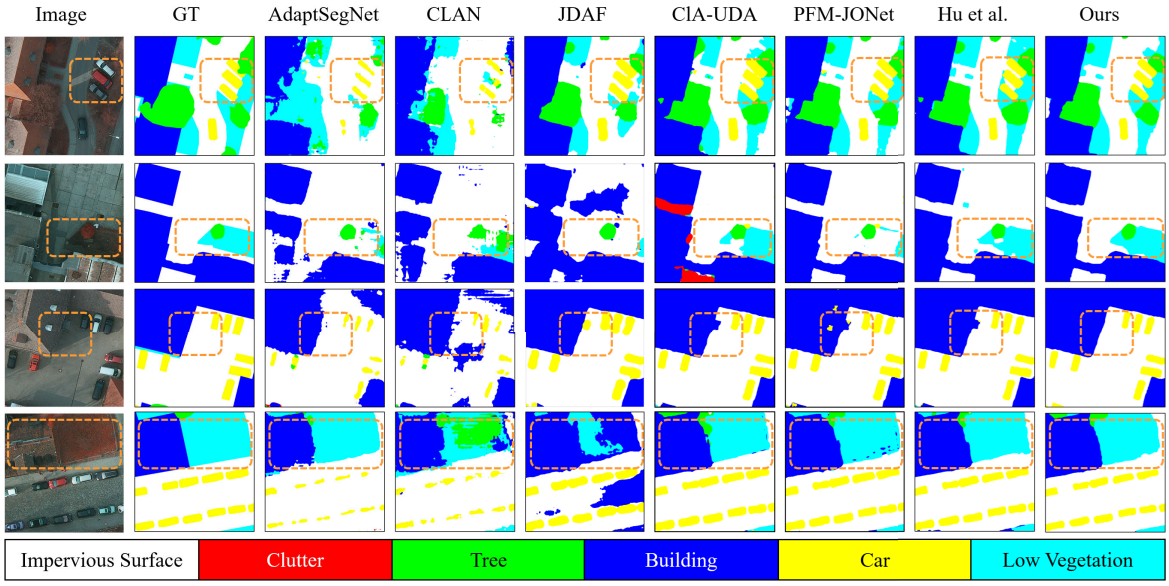


Fig. 7. Visual comparison on the Vaihingen → Potsdam task. From left to right: input images, ground-truth segmentation maps, results of state-of-the-art methods, and results of our method.

"Agriculture," "Forest," and "Water" classes, with fewer "Road" and "Building" samples. GMMDA achieves strong performance on several sample-rich categories, including a 7.66% improvement over the second-best method on the "Agriculture" class. It also maintains competitive performance on sample-scarce categories.

In the second row of Fig. 8, only our method correctly identifies the "Water" region (blue) beneath "Agriculture" (orange), whereas other methods misclassify this area. In the first, third, and fourth rows, our method preserves the outlines of "Building" (red) more completely, whereas other methods confuse parts of these regions with the "Background" (white). Notably, in the third row, the other methods fail to fully delineate the "Forest" region (green) surrounding "Building," whereas our method shows less confusion between the two categories.

4) *LoveDA Urban ⟶ LoveDA Rural Task:* The comparison results for the Urban→Rural task are presented in Table II (bottom). The urban subset mainly consists of "Road" and "Building," with fewer samples for "Agriculture" and "Water." The experimental results show that our method achieves the best overall performance and improves several challenging categories in complex remote-sensing scenarios. It also remains competitive on underrepresented categories, suggesting

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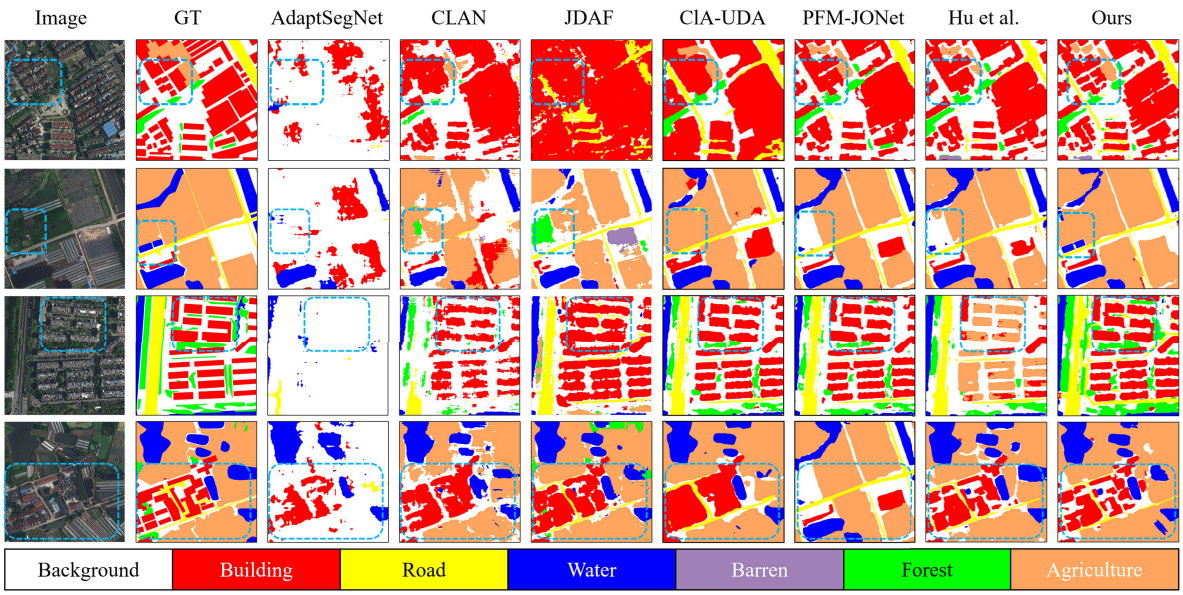


Fig. 8. Visual comparison on the Rural → Urban task. From left to right: input images, ground-truth segmentation maps, results of state-of-the-art methods, and results of our method.

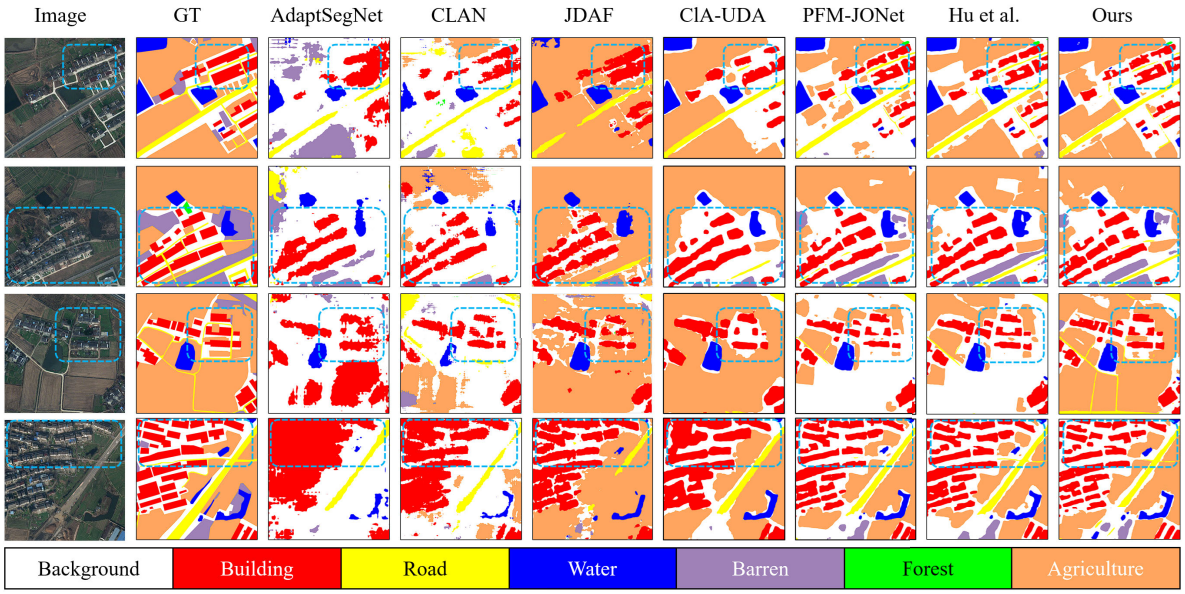


Fig. 9. Visual comparison on the Urban → Rural task. From left to right: input images, ground-truth segmentation maps, results of state-of-the-art methods, and results of our method.

that GMMDA can maintain stable performance under class imbalance and complex backgrounds.

Fig. 9 provides a visual comparison of segmentation results from different models. In the first and third rows, our method preserves the "Agriculture" regions more completely, a category with fewer source-domain samples. In the second row, our method correctly identifies multiple separated "Building" regions without merging them into other categories. In the fourth row, our method better preserves the boundaries among "Agriculture," "Building," and "Road," with less confusion across adjacent categories.

Overall, GMMDA obtains the best mIoU among the compared methods across the four tasks. In terms of per-class IoU, it improves several categories and yields a relatively balanced performance across classes. In particular, for categories with substantial appearance variation (e.g., car and building) and those prone to boundary ambiguity (e.g., tree and low veg.), GMMDA yields more stable predictions and produces more spatially coherent segmentation maps.

C. Ablation Study

In this section, we conduct ablation experiments on individual components of the proposed network to validate the effectiveness of each module. All modules are evaluated on the Potsdam→Vaihingen and Rural→Urban tasks. The corresponding quantitative results are reported in Table III.

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TABLE III
ABLATION STUDY OF DIFFERENT COMPONENTS IN THE PROPOSED FRAMEWORK

Task	Source Only	EMA-GMM	DMM	CCL	CDA	mIoU (%)	Gain
Potsdam ↓ Vaihingen	✓					39.22	-
		✓				55.40	↑16.18
		✓	✓			56.81	↑17.59
		✓		✓		61.71	↑22.49
		✓			✓	60.30	↑21.08
		✓	✓	✓		61.89	↑22.67
		✓	✓		✓	62.60	↑23.38
		✓		✓	✓	63.91	↑24.69
		✓	✓	✓	✓	**64.32**	**↑25.10**
Rural ↓ Urban	✓					36.45	-
		✓				47.80	↑11.35
		✓	✓			49.31	↑12.86
		✓		✓		52.27	↑15.82
		✓			✓	51.86	↑15.41
		✓	✓	✓		53.04	↑16.59
		✓	✓		✓	52.77	↑16.32
		✓		✓	✓	53.82	↑17.37
		✓	✓	✓	✓	**54.08**	**↑17.63**

1) *Source Only:* We employ SegFormer [44] as the baseline. Under the Potsdam → Vaihingen setting, the model achieves an mIoU of 39.22%. In the Rural → Urban setting, the mIoU reaches 36.45%.
2) *EMA-GMM:* Based on a SegFormer self-training framework, we replace the traditional discriminative classifier with our generative classifier and introduce the commonly used exponential moving average (EMA) network and image augmentation strategies.
3) *DMM:* A DMM that stores representative source-domain pixel features during training, thereby enriching the data for GMM modeling and providing more stable source components for CCL and CDA.
4) *CCL:* CCL that encourages target-domain features to align with their best-matched Gaussian components while remaining separable from other intraclass subclusters.
5) *CDA:* CDA that calibrates cross-domain structural discrepancies at the component level by jointly constraining distribution means and principal-subspace directions.

As shown in Table III, each module contributes to the overall system performance, and the best results are obtained when they are used together. First, in the Potsdam→Vaihingen and Rural→Urban adaptation tasks, the EMA-GMM framework outperforms the baseline by 16.18% and 11.35%, respectively.

Furthermore, we conduct an independent analysis to evaluate the effect of classifier replacement on performance in the two UDA tasks. To ensure a fair assessment of the generalization ability of the model and reduce the potential bias introduced by individual random seeds, we use three random seeds. As illustrated in Fig. 10, the blue curve corresponds to the model with a discriminative classifier, whereas the orange curve corresponds to the same model with a GMM-based generative classifier. The results show that, during the early stages of training, the discriminative classifier exhibits stronger learning capability, since GMM modeling requires sufficiently representative features to perform effectively.

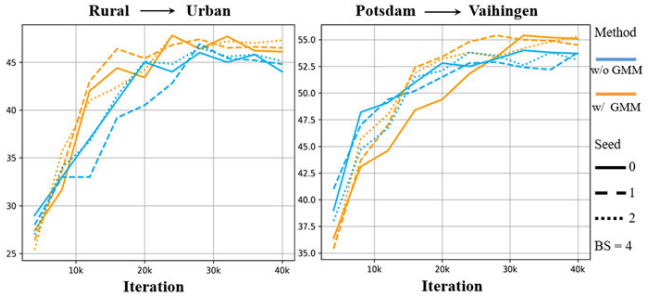


Fig. 10. Comparison of UDA performance of the base framework with and without the GMM classifier. All curves are trained with a batch size of 4.

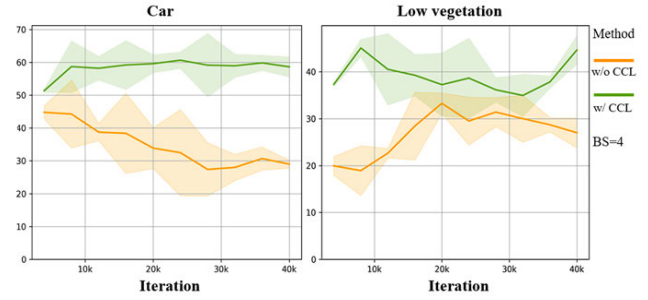


Fig. 11. UDA performance on the hard-to-adapt classes "Car" and "Low vegetation" with and without the CCL module. All curves are trained with a batch size of 4.

However, as self-training proceeds, the GMM-based classifier gradually shows better segmentation performance than the discriminative baseline under all three random seeds, suggesting the effectiveness of replacing the classifier within the current framework.

Compared to the EMA-GMM model, introducing CCL improves performance by 6.31% and 4.47%, respectively.

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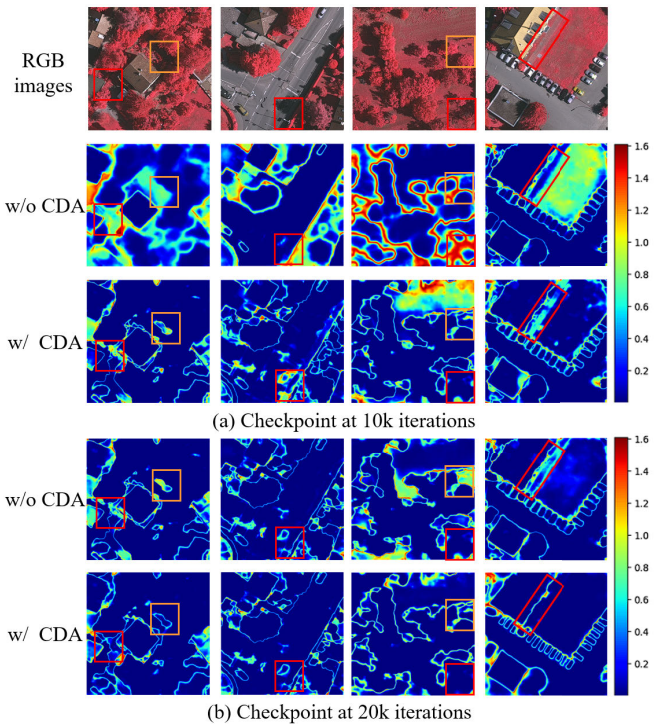


Fig. 12. Predictive-entropy maps with and without the CDA module. The first row presents red, green, and blue (RGB) image crops, and the following rows show entropy maps at the same training checkpoints (10k and 20k iterations). Red indicates higher entropy and blue indicates lower entropy under a shared color scale. Boxes highlight representative transition areas.

Notably, CCL aligns target-domain features through componentwise contrast learning, in which each pixel is associated with its most probable Gaussian component under the predicted class. This design encourages alignment at the intraclass component level rather than relying only on a single class vector. To further evaluate the effectiveness of the CCL module, we report the mIoU of the "Car" and "Low veg." classes, which exhibit the most pronounced changes after the introduction of CCL. As shown in Fig. 11, the green curve (with CCL) shows clear performance gains for both "Car" and "Low veg." and follows a more stable upward trend during training. In contrast, the orange curve (without CCL) remains lower and fluctuates more noticeably, suggesting that CCL can improve training stability on visually challenging categories within the current framework.

CDA improves the EMA-GMM model by 4.90% and 4.06%. To reduce the confounding effect of training progress, we compare predictive-entropy maps for models with and without CDA at iteration-matched checkpoints. As shown in Fig. 12, CDA noticeably lowers predictive entropy in ambiguous transition regions (boxed), indicating lower uncertainty and more consistent boundary predictions. Under domain shift, such regions are particularly sensitive to class-conditional structural mismatch, such as discrepancies in component centers and geometry. By jointly regularizing component means and principal-subspace directions, CDA provides a structure-aware alignment signal that helps alleviate this mismatch.

Moreover, the experimental results suggest that DMM can further support the other modules. By providing representative source-domain samples, DMM helps improve the reliability of GMM estimation and offers more stable source components for CCL and CDA during training.

TABLE IV

EFFECT OF THE NUMBER OF GAUSSIAN COMPONENTS PER CLASS. THE BEST RESULTS ARE MARKED IN RED

M	Potsdam ↓ Vaihingen	Vaihingen ↓ Potsdam	Rural ↓ Urban	Urban ↓ Rural	Average
1	57.01	60.20	51.60	41.42	52.56
2	59.33	60.77	52.88	43.28	54.07
3	62.70	61.16	53.54	44.68	55.52
4	63.69	61.21	54.08	45.88	56.22
5	64.32	60.55	53.41	46.33	56.15
6	62.57	58.72	53.27	45.74	55.08
7	61.69	57.43	52.98	45.71	54.45
8	60.88	56.59	52.60	45.60	53.92

D. Sensitivity Analysis and Feature Visualization

1) Effect of the Number of Gaussian Components: In GMMDA, the number of Gaussian components per class, denoted by M, controls the tradeoff between representation capacity and estimation stability. As shown in Table IV, increasing M from 1 to a moderate value consistently improves performance, indicating that a single Gaussian component is insufficient to capture multipeaked intraclass feature distributions. Across the four tasks, $M = 4$ and $M = 5$ achieve the best average performance, whereas larger values lead to a consistent performance decline.

This trend indicates that using more components is not always beneficial. When M is too small, different intraclass modes may be forced into the same component, leading to underfitting of complex class-conditional distributions. In contrast, when M is too large, each component receives fewer effective samples, making the estimation of component means and covariances less stable and increasing the risk of overparameterization. This issue is particularly relevant to GMMDA because the estimated covariance further affects the Mahalanobis-distance-based CCL and the principal-subspace-based CDA.

Table IV provides a sensitivity analysis rather than a target-label-based hyperparameter selection procedure. Based on the average trend, we recommend a moderate value, such as $M = 4$ or $M = 5$, as a practical default range. For a new dataset without target-domain annotations, M can be selected according to source-domain intraclass diversity and label-free training statistics, such as component occupancy and effective sample size. A reasonable choice of M should avoid both underfitting caused by too few components and unstable or poorly occupied components caused by too many components.

2) Effect of the Loss Weights: To analyze the sensitivity of the loss weights, we conduct a sensitivity analysis on the Potsdam → Vaihingen task. We systematically evaluate a grid of weight combinations and record the resulting mIoU. To better visualize the mIoU differences, higher scores are shown

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Fig. 13. Ablation study of the weights α and β. Color intensity indicates model performance, with darker colors representing better performance under different weight combinations.

in darker colors, as shown in Fig. 13. When α and β are small, the contributions of CCL and CDA are limited, and the overall loss is dominated by $\mathcal{L}_{\text{CE}}^{S}$ and $\mathcal{L}_{\text{CE}}^{T}$, resulting in only modest performance improvement. Conversely, when α and β are too large, they may interfere with the model's optimization process, leading to performance degradation. When $\alpha = 0.3$ and $\beta = 0.4$, the two loss terms are well balanced, and the model achieves the best mIoU.

3) Model Complexity and Efficiency: As a generative alternative, GMMDA introduces additional computations, including componentwise likelihood evaluation, Mahalanobis-distance computation, and covariance-related parameter estimation. To evaluate its practical overhead, we first compare GMMDA with a SegFormer-based self-training UDA baseline under the same backbone, training protocol, input crop size, hardware, and inference setting. This controlled comparison directly reflects the additional complexity introduced by the proposed GMM-based modeling.

We report two complementary efficiency comparisons. For the controlled comparison with the SegFormer-ST baseline, we report training throughput, training memory, inference frames per second (FPS), inference memory, and model parameters under the same training and inference settings. For reference comparisons with representative UDA methods, we report model parameters and inference FPS, because these methods may use different backbones, architectures, and implementation protocols, making training throughput and memory less directly comparable.

As shown in Table V, GMMDA introduces moderate training overhead but only limited inference overhead compared with the SegFormer-ST baseline. The training throughput decreases from 3.91 to 2.77 img/s, and the peak training memory increases from 8.1 to 11.4 GB. The additional training cost mainly comes from the memory-assisted online EM update, covariance estimation, CCL, and CDA.

During inference, the training-time modules, including the DMM, online EM update, CCL, and CDA, are discarded. Therefore, the inference-stage overhead mainly comes from the learned GMM classifier. The inference FPS decreases only slightly from 12.7 to 12.0, and the inference memory increases from 2.10 to 2.27 GB. The number of parameters also increases only marginally from 84.6M to 84.8M, since the additional Gaussian component parameters are lightweight compared with the SegFormer backbone.

TABLE V

EFFICIENCY COMPARISON UNDER THE SAME UDA TRAINING PROTOCOL. RATIO DENOTES GMMDA/SEGFORMER-ST, AND TRAINING THROUGHPUT IS MEASURED IN IMAGES PER SECOND

Metric	SegFormer-ST	GMMDA	Ratio
Protocol	UDA	UDA	-
Train throughput	3.91 img/s	2.77 img/s	0.71×
Train memory	8.1 GB	11.4 GB	1.41×
Inference FPS	12.7	12.0	0.94×
Inference memory	2.10 GB	2.27 GB	1.08×
Parameters	84.6M	84.8M	1.00×

TABLE VI

REFERENCE COMPARISON OF MODEL COMPLEXITY AND INFERENCE EFFICIENCY. FPS IS MEASURED WITHOUT TEST-TIME AUGMENTATION

Method	Backbone/ Architecture	Input	Total Params	Trainable Params	FPS
SegFormer-ST	MiT-B5	512 × 512	84.6M	84.6M	12.7
AdaptSegNet	ResNet-101	512 × 512	42.8M	42.8M	19.7
CLAN	ResNet-101	512 × 512	43.4M	43.4M	17.5
PFM-JONet	MiT-B5 + SAM	512 × 512	233.5M	138.3M	9.5
Hu et al.	MiT-B5	512 × 512	88.5M	88.5M	11.4
ABFNet	Res2Net-101	512 × 512	213.6M	213.6M	7.5
GMMDA	MiT-B5	512 × 512	84.8M	84.8M	12.0

TABLE VII

EFFECT OF THE CONFIDENCE THRESHOLD t ON THE LOVEDA RURAL→URBAN TASK. L_{CE}^{T} DENOTES THE TARGET-DOMAIN SELF-TRAINING LOSS. THE BEST MIOU RESULT IS MARKED IN RED

t	0.5	0.7	0.8	0.9	0.99
L_{CE}^{T}	0.2300	0.2170	0.1772	0.1636	0.1403
mIoU (%)	52.16	53.27	53.41	54.08	53.77

We further report representative UDA methods as reference comparisons in Table VI. GMMDA has a comparable parameter scale and inference speed to methods using MiT-B5-based architectures, while being more lightweight than hybrid or multimodal architectures such as PFM-JONet and ABFNet. This comparison is intended as a reference rather than a strictly controlled comparison.

4) Effect of the Confidence Threshold t: The confidence threshold t controls which target-domain pixels are retained for self-training. Only pixels whose maximum predicted probability is no lower than t are selected by the confidence mask and contribute to the target-domain self-training loss L_{CE}^{T}. In all experiments, we set $t = 0.9$.

We conduct a sensitivity analysis of t on the LoveDA Rural→Urban task. As shown in Table VII, different threshold values affect both the target-domain self-training loss and the

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TABLE VIII
COMPARISON WITH THE CENTROID-ONLY MULTIPROTOTYPE K-MEANS VARIANT. TRAINING THROUGHPUT AND INFERENCE FPS ARE REPORTED RELATIVE TO THE K-MEANS VARIANT; VALUES BELOW 1.00× INDICATE LOWER SPEED

Method	Representation	Potsdam→Vaihingen	Rural→Urban	Train Throughput	Infer. FPS
Multi-prototype K-Means	Mean only	62.86	52.27	1.00×	1.00×
GMMDA	Mean + covariance	64.32	54.08	0.92×	0.98×

final segmentation accuracy. A lower threshold retains more target-domain pixels but may introduce noisier pseudo-labels, leading to less reliable supervision. As t increases from 0.5 to 0.9, the selected pseudo-labels become more reliable, and the mIoU improves accordingly. However, an excessively high threshold retains only a small number of easy pixels, which weakens effective target-domain supervision and limits further improvement.

It should be noted that L_{CE}^{T} is computed only over the selected target-domain pixels, and different thresholds correspond to different pseudo-label sets. Therefore, the self-training loss alone cannot fully reflect adaptation quality and should be interpreted together with mIoU. Although $t = 0.99$ yields the lowest L_{CE}^{T}, its mIoU is slightly lower than that of $t = 0.9$, suggesting that an overly strict threshold may reduce target-domain coverage. Overall, $t = 0.9$ achieves the best segmentation accuracy among the tested settings and provides a good balance between pseudo-label reliability and target-domain supervision strength.

5) Comparison With Centroid-Based Multiprototype Modeling: To examine whether the improvement of GMMDA comes only from using multiple component centers, we construct a centroid-only multiprototype K-means variant. For a fair comparison, this variant uses the same backbone, number of components, DMM, and self-training setting as GMMDA. It represents each component only with a centroid, uses Euclidean distance for component matching and CCL, and removes the principal-subspace alignment term in CDA.

As shown in Table VIII, the multiprototype K-means variant achieves competitive results, confirming the benefit of multiprototype modeling. However, GMMDA further improves mIoU from 62.86% to 64.32% on the Potsdam → Vaihingen task and from 52.27% to 54.08% on the Rural → Urban task. This suggests that the advantage of GMMDA comes not only from multiple component centers but also from covariance-aware component modeling. By modeling both the mean and covariance of each component, GMMDA captures the scale and directional variance of intraclass subdistributions, which benefits Mahalanobis-distance-based component matching in CCL and principal-subspace-based distribution alignment in CDA.

In terms of computational overhead, GMMDA retains 0.92× training throughput and 0.98× inference FPS compared with the centroid-only K-means variant. These results show that GMMDA achieves consistent performance gains with moderate additional training cost and marginal inference overhead, providing a favorable performance-cost tradeoff in our framework.

6) Effect of DMM Capacity: To isolate the influence of DMM capacity, we conduct a sensitivity analysis under the EMA-GMM + DMM setting. This follows the module-wise ablation setting, where DMM is introduced on top of EMA-GMM before adding CCL and CDA. In this way, the effect of memory capacity can be more directly attributed to memory-assisted GMM estimation.

TABLE IX
SENSITIVITY ANALYSIS OF DMM CAPACITY. THE CAPACITY DENOTES THE MAXIMUM NUMBER OF STORED FEATURES IN EACH CLASS-COMPONENT QUEUE

Capacity	Potsdam ↓ Vaihingen	Rural ↓ Urban
0	55.40	47.80
8K	56.11	48.42
16K	56.55	48.99
32K	56.81	49.31
48K	56.80	49.33

As shown in Table IX, increasing the DMM capacity improves performance when the capacity is small, indicating that historical features provide more stable statistics for estimating class-component GMM parameters. The performance saturates around 32K, and further increasing the capacity to 48K brings only marginal improvement. Therefore, we set the default capacity to 32K to balance componentwise statistical stability and memory cost.

TABLE X
SENSITIVITY ANALYSIS OF THE NUMBER OF SPARSELY SAMPLED FEATURES PER CLASS AT EACH MEMORY UPDATE

Samples per class	Potsdam ↓ Vaihingen	Rural ↓ Urban
25	56.12	48.67
50	56.69	49.06
100	56.81	49.31
200	56.76	49.26

7) Effect of the Number of Sparsely Sampled Features: We further evaluate the effect of the number of sparsely sampled features per class at each memory update. As shown in Table X, increasing the sampling number from 25 to 100 improves performance, indicating that sufficient sampled features are needed for stable componentwise GMM estimation. Further increasing the sampling number to 200 brings no additional improvement, suggesting that the benefit becomes saturated and that more neighboring pixels may introduce redundant information. Therefore, we set the default sampling number to 100 to balance feature diversity and update efficiency.

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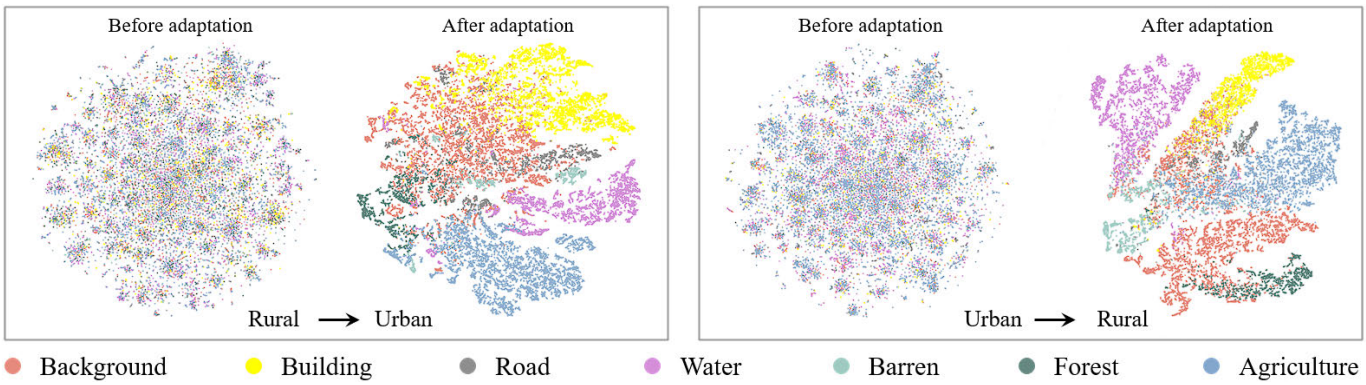


Fig. 14. t-SNE visualization of feature distributions before and after domain adaptation for Rural ⟷ Urban tasks.

TABLE XI
SENSITIVITY ANALYSIS OF THE LOW-RANK SUBSPACE DIMENSION r

Subspace dimension r	Potsdam ↓ Vaihingen	Rural ↓ Urban
2	62.54	53.42
4	63.61	53.86
8	64.32	54.08
16	64.28	54.01
32	64.01	53.85

TABLE XII
ABLATION STUDY OF RCS FOR LONG-TAILED CLASS ADAPTATION. RARE MIOU IS COMPUTED OVER THE THREE LEAST FREQUENT SOURCE-DOMAIN CLASSES IN EACH TASK

Metric	w/o RCS	w/ RCS	Gain
Potsdam → Vaihingen			
Rare mIoU	62.41	64.18	+1.77
mIoU	63.47	64.32	+0.85
Rural → Urban			
Rare mIoU	48.36	50.82	+2.46
mIoU	52.82	54.08	+1.26

8) Effect of the Low-Rank Subspace Dimension: The low-rank subspace dimension r controls the number of principal directions retained in the low-rank-plus-diagonal covariance representation. As shown in Table XI, increasing r from 2 to 8 consistently improves performance, indicating that retaining sufficient principal directions benefits covariance-aware component modeling. A very small r preserves only limited directional information and may not sufficiently describe intracomponent variations. However, when r is further increased to 16 or 32, the performance saturates and slightly decreases. This may be because directions associated with small eigenvalues tend to capture noisy or unstable variations, which can make covariance and subspace estimation less robust.

Therefore, we set $r = 8$ as the default value because it achieves the best performance among the tested settings while maintaining a compact and stable covariance representation.

9) Effect of RCS: We evaluate the contribution of RCS to long-tailed class adaptation by replacing the default RCS sampling with ordinary random source sampling while keeping all other settings unchanged. Rare mIoU is computed over the three evaluated classes with the lowest source-domain pixel frequencies in each task. Specifically, the rare classes are tree, car, and clutter for Potsdam → Vaihingen, and road, building, and barren for Rural → Urban.

As shown in Table XII, RCS improves Rare mIoU by 1.77% and 2.46% on Potsdam → Vaihingen and Rural → Urban, respectively. These gains are larger than the corresponding overall mIoU improvements of 0.85% and 1.26%. This indicates that RCS mainly benefits underrepresented categories by increasing their occurrence frequency in source-domain mini-batches, while also improving overall adaptation performance.

10) Visualization of Feature Distribution: To further illustrate the effectiveness of the proposed method, we use t-SNE [54] to visualize the feature distributions on the LoveDA dataset, as shown in Fig. 14. The visualization suggests that the features learned by our method are more compact and more separable across categories, with clearer boundaries between clusters corresponding to different categories, each marked by a distinct color. This improved compactness and separability are consistent with the better segmentation performance observed in the quantitative results. In contrast, the features produced by the source-only model are more dispersed and exhibit substantial overlap, with less distinct boundaries between categories. This weaker separation is also consistent with its lower segmentation performance and reflects the pronounced domain gap between the source and target domains, which the source-only baseline fails to mitigate effectively.

V. CONCLUSION

We propose GMMDA, a generative UDA method for semantic segmentation that models class-conditional distributions with class-wise Gaussian mixtures and predicts labels via Bayesian inference. GMMDA updates the mixture parameters through online momentum Sinkhorn EM and optimizes the network through end-to-end back-propagation. By modeling each class with multiple Gaussian components, GMMDA is designed to characterize intraclass variability and provides

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explicit component-level structure. Building on this structure, CCL performs responsibility-guided component matching for target-domain features to avoid coarse-grained attraction from a single-vector representation. CDA further calibrates each source component with the corresponding source-guided target local component statistics by jointly aligning their means and principal-subspace directions, thereby reducing structural mismatch. Extensive experiments on multiple benchmarks demonstrate that GMMDA achieves competitive performance. We hope that this work encourages further exploration of generative, distribution-aware modeling as a useful complement to existing discriminative methods for robust cross-domain semantic segmentation.

In future work, adaptive selection of the number of Gaussian components could be further explored. Instead of using a fixed M for all classes, class-specific component numbers may be estimated using label-free training statistics, such as component occupancy, assignment entropy, effective sample size, and covariance stability. Such statistics could support occupancy-based pruning and split-and-merge strategies, which may better match the varying complexity of different semantic classes without relying on target-domain annotations.

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